SANITARY

TOPIC TWO - SOIL AND WASTE WATER APPLIANCES USED IN BUILDINGS

INTRODUCTION.

In buildings, many types of sanitary appliances are required. They fulfil a variety of specialized functions, which are connected with drainage and disposal of sewage and refuse from buildings. They can be divided into waste appliance and soil appliances.

Soil water/ black water is wastewater containing human waste from toilets, urinals, or budgets.

Wastewater/grey water is water from sinks, showers, baths, and washing machines and contains dirt, grease, soap, or food particles.

Definition of terms

1. **Sanitary Appliances or fixtures**- are those sanitary objects or appliances that are fixed or attached. They are commonly referred to as sanitary furniture. They are static, immovable, and motionless. Sanitary fixtures are sanitary wares or *furniture attached and installed in different parts of a building and they are permanently connected to a plumbing system to be used for the reception and discharge of water or waste*. Examples of sanitary fixtures include water closet, lavatory or washbasin, urinal, floor drain, flushing cisterns, showers, faucets, hose bibs, bathtubs, dishwashers, sinks, laundry sink, drinking fountain, bidet to mention a few.
2. **Sanitary fittings**- also known as hygienic fittings or sanitary clamps are those objects or appliances used to seal, clamp and fasten two or more separate ends of sanitary fixtures from any crevices or gaps to maintain a high hygienic standard. In other words, sanitary fittings are used to create closures, linkages, and a secure connection between sanitary fixtures so as not to create any leaking site or gap that can harbor or lead to exposure of bacteria and germs. Examples adapter, coupling, sleeve, union, cap, plug, elbow, tee, wye, cross, reducer etc.
3. **Soil System** - The soil system refers to the network of pipes and fittings responsible for collecting and conveying wastewater from sanitary fixtures to the sewer or sewage treatment system.
4. **Trap** - A device designed to retain a small amount of water to prevent the passage of sewer gases into living spaces while allowing wastewater to flow freely.

5. **Sewerage System** - The entire network of pipes, pumps, and other infrastructure designed to collect and transport wastewater from various sources to treatment plants or disposal points.
6. **Backflow Prevention** - The installation of devices or methods to prevent the reversal of flow in a drainage system which could contaminate the potable water supply.
7. **Grease Trap** - A device installed in commercial kitchens to capture and retain grease and solids before they enter the wastewater disposal system, preventing blockages and reducing environmental impact.
8. **Effluent** - The treated or untreated liquid discharge from a wastewater treatment system, septic tank, or other plumbing appliance
9. **Water seal** A trap water seal is a small amount of water retained in the U-shaped bend of a plumbing trap (like P-traps or S-traps) under fixtures. It acts as a hygienic, airtight barrier, preventing noxious sewer gases, foul odors, bacteria, and pests from entering the building while allowing wastewater to flow freely. Key details about trap water seals include:

- **Purpose:** The primary function is to block methane, hydrogen sulfide, and other hazardous gases from entering living spaces.
- **Location:** Found in all plumbing fixtures, including sinks, showers, toilets, and floor drains.
- **Mechanism:** The U-bend retains water after a fixture is used, creating a physical barrier.
- **Potential Failures:** The seal can be lost through evaporation in unused drains, backpressure, or siphoning (suction).
- **Solutions:** If a trap dries out, water must be added, or a trap seal primer (a device that adds water) or a specialized "trap seal" (a one-way valve) can be used to maintain the barrier.

Types of traps commonly used to create this seal include P-traps (common in sinks), S-traps, and bottle traps.

1. **SANITARY APPLIANCES.**

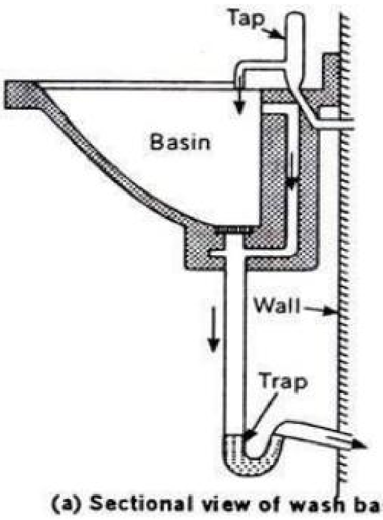
There are six main types of sanitary fixtures

a) Washbasin(LAV-US)

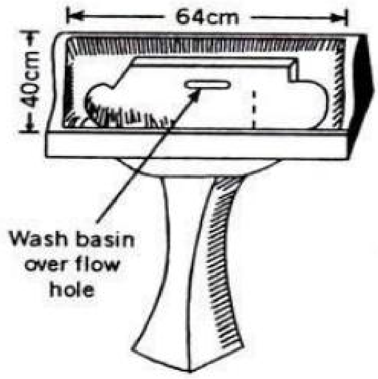
The standard basin consists of a bowl, weir overflow and holes for taps. The basin may be supported on brackets screwed to the wall or a pedestal which conceals the pipework. It can also be top mounted on a worktable

For best functionality, a washbasin should:

- Have slotted overflow hole.
- Have all their internal angles designed so as to facilitate cleaning.
- Should be provided with a circular waste hole in the bottom.
- The basins are provided with an integral soap holder recess which drains into the bowl.
- For holding water in the bowl there are provided with tapering rubber plugs, which can be fitted in the outlet.



(a) Sectional view of wash basin.

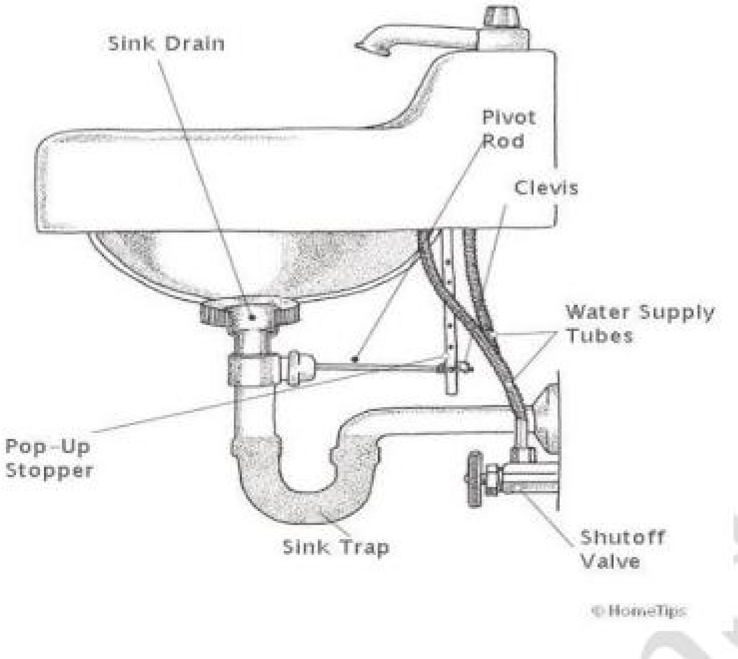


(b) Isometric view of wash basin.

b) Sinks

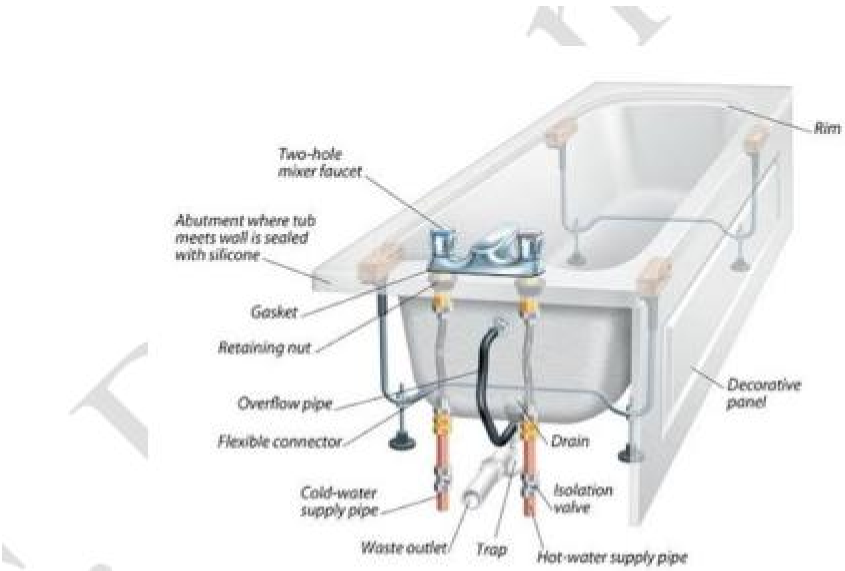
These are rectangular shallow receptacles suitable for kitchens or laboratory The floor of the sink is given a slope towards the waste outlet. The sinks are provided with circular waste hole. All the kitchen sinks are provided with a draining board which is fixed on the right of the user. However, there are also double bowl double drain sinks that have draining boards on each sides. Overflow slots are also provided in some sinks.

✓ Used for washing hands, dishes, and other cleaning tasks, sinks come in various materials like stainless steel, porcelain, and cast iron. They can be single or double bowl, with additional features



c) Bath tabs.

The bath should also be provided with one over-flow pipe to take excessive water. The waste pipe of bath is provided with a trap, to prevent the foul gases from entering in the bath room.



d) Flushing Cisterns:

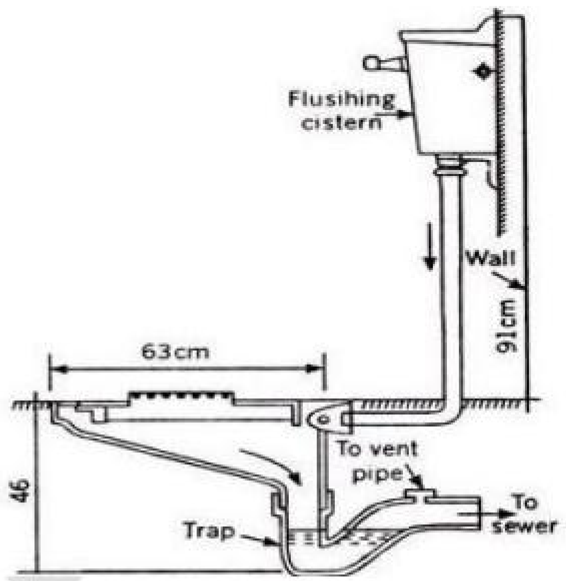
These are used for flushing water closets and urinals after use. The flushing cisterns are provided with inlet pipe, over-flow pipe and automatic closing float ball valve.

e) **Water-Closet**

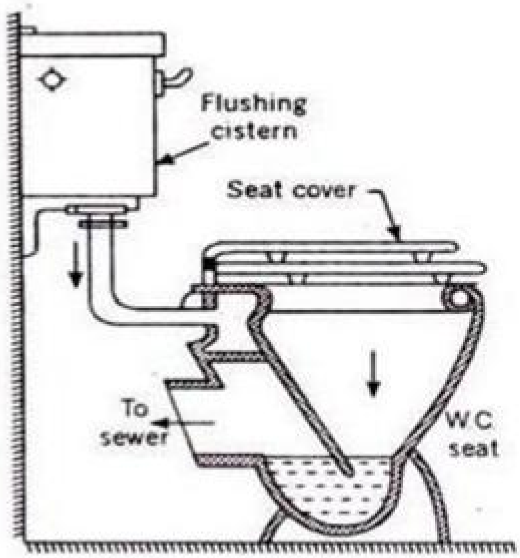
This is a sanitary appliance to receive the human excreta directly and is connected to thesoil pipe by means of a trap.

The water closets are classified as follows:

i. Squatting type or Indian type



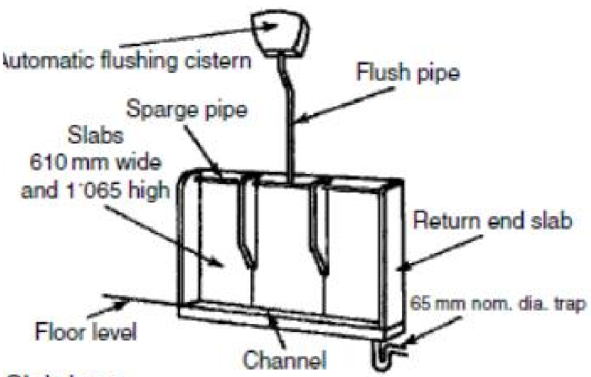
ii. Wash-down, Pedestral or European type



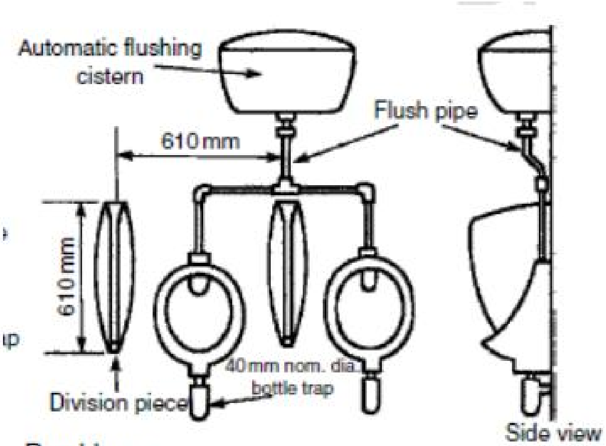
f) Urinals:

Used to collect urine. The inside surface is regular and smooth for ensuring efficient flushing. The bottom of the urinal is provided with sufficient slope from the front towards the outlet for efficient drainage of the urinal.

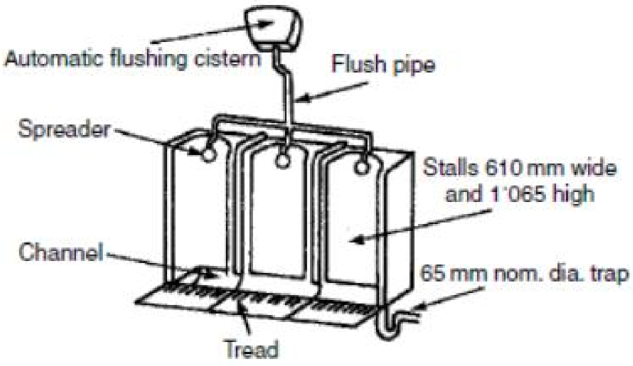
Urinals are washed at intervals of 20 minutes by means of an automatic flushing cistern discharging 4.5 litres of water per bowl of 610 mm of slab/stall width. The water supply to the cistern should be isolated by a motorized valve on a time control, to shut off when the building is not occupied.



Slab type



Bowl type



Stall type

The slab and stall type urinals are manufactured either as a urinal or as a range of two or more and are used in public places such as cinema houses, restaurants, railway stations, offices, etc.

The squatting plate urinals are mostly used in ladies' lavatories and are on piece construction
A bidet is a sanitary appliance used for personal hygiene after using the toilet. It typically involves spraying water onto the genital and anal area.

Features:

- Elongated oval or D-shaped bowl to comfortably accommodate different body types.
- Adjustable nozzle with various spray patterns and water pressure settings for personalized cleansing.
- Optional feature for added comfort, especially in cold climates.
- Optional feature for hygienic and comfortable drying.
- Smooth, non-porous surfaces for efficient cleaning and preventing bacteria growth.

h) Faucets

These control the flow of water from pipes to sinks, bathtubs, showers, and bidets. They come in various styles and functionalities, including single-handle, double-handle, touchless, and sensor-activated options.

2) SANITARY FITTINGS

Sanitary fittings can be classified as follows:

a) The connection types such as ball and sleeve fittings, barbed fittings, cam-lock fittings,

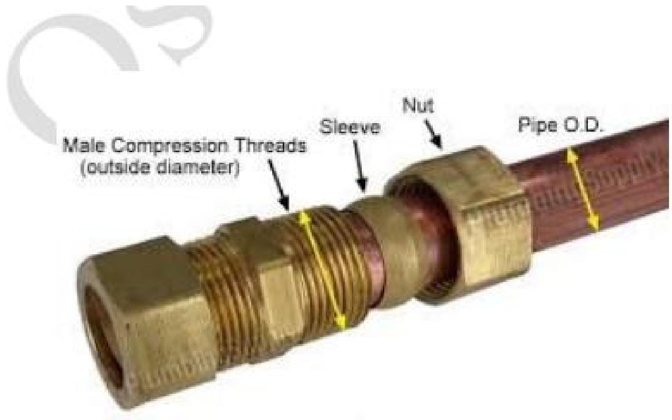
compression fittings, crimp fittings, end fittings, flanges, threaded fittings

Ball and sleeve fittings-Connects an outer sleeve to an inner (ball) fitting. The sleeve retracts to connect and disconnect the two ends of the fitting.

Barbed fittings-Connects hose flexible tubing via a barbed end with a tapered stub and ridges inserted into the vessel. They are best suited for low pressure applications, since they do not provide a strong seal

Cam-lock fittings-Connected using tabs which fold down into the receiver to lock the fitting in place after insertion. They are used in many heavy-duty applications such as fire hose and sludge"sewage pumping

Compression fittings-Connect vessels using compression on a gasket, ring, or ferrule



Crimp fittings-involves placing a vessel over a tubular end and crimping against it with a sleeve or crimp socket. These fittings typically require crimping tools to make the connections

Plain ends are fittings which allow vessels to be connected by adhesive, solder, or other forms

Flanges- This is a pipe fitting used to connect pipes, pumps, valves and other components to form a full-fledged piping system. They come with a flexibility of easily cleaning or inspecting the whole system from within. They are fixed to the pipes using welding, threading or screwing techniques and then finally sealed with the help of bolts. They are used in residential pump systems and majorly for industrial purposes.

b) The function of such fitting such as fittings which extends or terminate (for example, adapter, coupling, sleeve, union, cap, plug etc.)

ADAPTERS-These are connected to pipes to either increase their lengths or if pipes do not have appropriate ends. These pipe fittings make the ends of the pipe either male or female threaded as per the need. This permits unlike pipes to be connected without any need of extensive setup. They are mostly used for PVC and copper pipes

COUPLINGS-This is a pipe fitting used to stop leakages in broken or damaged pipes. The pipes to be connected should be of the same diameter.

UNIONS-This type of pipe fitting is almost similar to coupling in terms of functions, with a difference in ease of removal i.e. a union can be removed easily any time while the coupling cannot.

CAPS & PLUGS-Both these pipe fittings are used to close the ends of the pipe either temporarily or permanently. The plugs are fitted inside the pipe and threaded to keep the pipe for future use. There are a good number of ways a cap can be applied to the pipe like soldering, glue, or threading depending on the material of the pipe.

c) Fittings which add or change directions (for example, elbow, tee, wye, cross)

ELBOWS-Such pipe fittings are used to change the direction of the flow. They are majorly available in two standard types- 90- and 45-degree angles owing to their high demand in plumbing. The 90 degrees elbow is used to connect hoses to water pumps, valves, and deck drains while the 45 degrees elbow is mostly used in water supply facilities

TEES-This T-shaped pipe fitting used in the plumbing system has one inlet and two outlets arranged at an angle of 90 degrees to the main pipe. This kind of fitting is used to connect the two pipes and make their flow direction as one. If all the three sides of this fitting are same in size, it is called equal tee, otherwise unequal tee

CROSS-This type of fitting contains four openings in all the four major directions. This fitting is adjoined to four pipes meeting at common point. There is either one inlet and three outlets or vice-versa to flow water or any other liquid in four different directions. These kinds of pipe fittings are commonly used in fire sprinkler systems.

WYES-Such type of pipe fittings are used in drainage systems and have a branch line at 45 degrees to keep the flow of water smooth. When the sanitary tees fail to work in a horizontal connection, such cases needs a wye.

d) Fittings which change vessel size (for example, reducer and olet),

REDUCERS-This pipe setting is used to reduce the flow size of the pipe from the bigger to smaller one. There are two kinds of reducers- concentric reducer and eccentric reducer. The former one is in the shape of a cone used for gradual reducing of the size of the pipe. The latter one has its one edge facing the mouth of the connecting pipe reducing the chances of air accumulation.

e) Fittings for regulating and controlling of flows (for example, gate valves, globe valves, angle valves, check valves – spring check valve and swing check valve – ball valves, metered valves, flow control valve, thermostatic valves, temperature relief valve, pressure reducing valve, flushometer, sensor operated valves),

f) Traps for preventing foul gas from sewers to back flow (for example, floor traps, gully traps, intercepting traps) etcetera

◆ **P-Trap**

- **Shape looks like the letter P**

- **Used where pipe goes through the wall**
- **Most common in modern plumbing**

◆ **S-Trap**

- **Shape looks like S**
- **Used where pipe goes through the floor**
- **Can cause self-siphonage (water seal loss)**

◆ **Q-Trap**

- **Similar to P-trap but with a slightly different bend**
- **Less common in practice**

☞ **These are not separate systems—just different geometries.**

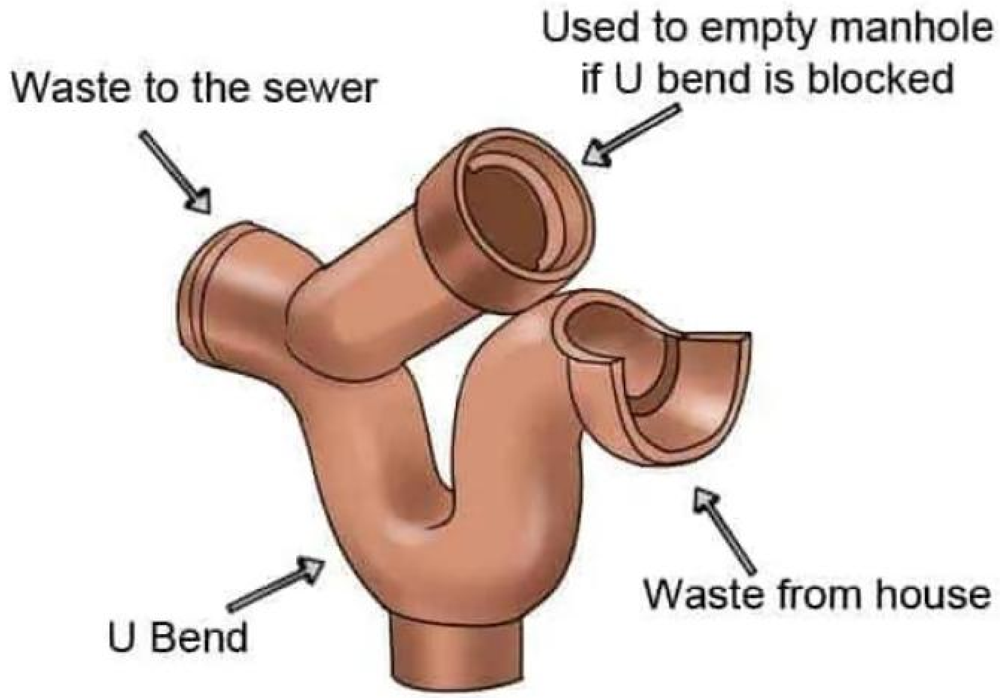
Gully Trap-These plumbing traps are built external to the building to carry wastewater released from sinks, washbasins, restrooms, and so on, and are associated with the most nearby building drain or sewers so that foul gases from the sewer don't go to the house. These are profound seal traps. In Gully Traps profundity of the water seal ought to be at 50 mm at least. It additionally keeps cockroaches and different bugs from sewer lines to squander pipes carrying wastewater. Gully traps are available in the following three shapes: *P-trap, S-trap and Q-trap*

Floor traps-Is a fitting which receives waste water from showers, bathrooms & kitchens floors and transmits it to drainage pipes. It is also used to prevent foul gases from the sewer drains from entering the living spaces

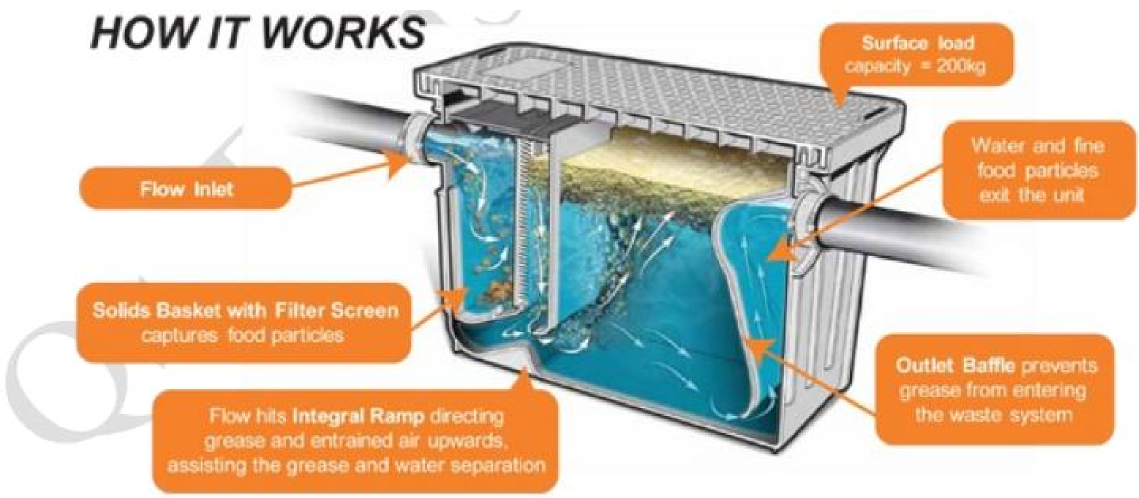
Bottle Trap-A bottle plumbing trap is given to get squander from the washbasin, kitchen sinks, and different machines where the apparatuses don't have an underlying trap

Intercepting Trap-Intercepting plumbing trap types are given into the Interceptor Manhole

(Interceptor Chamber). An Interceptor sewer vent is provided at the interference of building sewer and public sewer. That blocking type of trap is given to keep the foul gases from public sewers entering the building sewer by providing a water seal



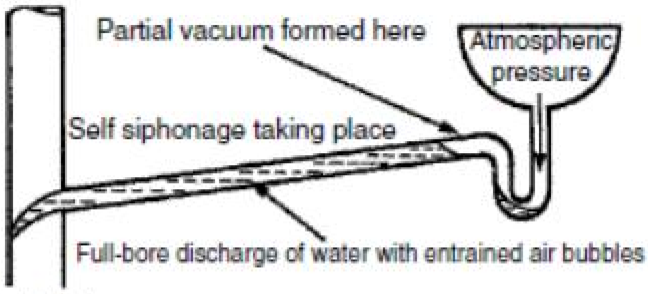
Grease Trap- A grease trap is introduced in the waste line from at least one installation to isolate grease from the fluid and hold it. This kind of trap is a gadget to gather the grease substance of waste and can be cleaned from the surface. This is mostly utilized in the food handling unit.



3) Loss of trap water seal.

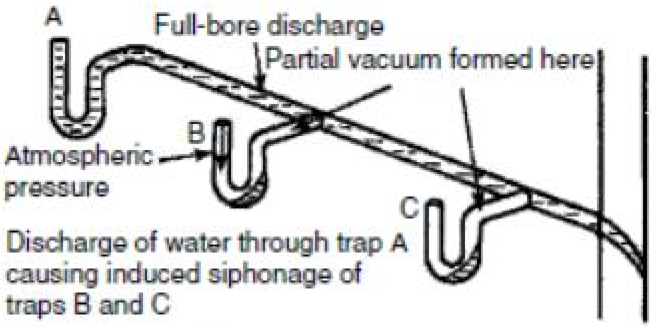
The most obvious cause of water seal loss is leakage due to defective fittings or poor workmanship. Otherwise, it may be caused by poor system design and/or installation:

• **Self-siphonage**: as an appliance discharges, the water fills the waste pipe and creates a vacuum to draw out the seal. Causes are a waste pipe that is too long, too steep or too small in diameter.



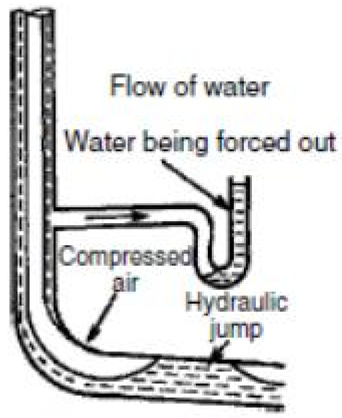
Self siphonage

• **Induced siphonage**: the discharge from one appliance draws out the seal in the trap of an adjacent appliance by creating a vacuum in that appliance's branch pipe. Causes are the same as for self-siphonage, but most commonly a shared waste pipe that is undersized. Discharge into inadequately sized stacks can have the same effect on waste branch Appliances



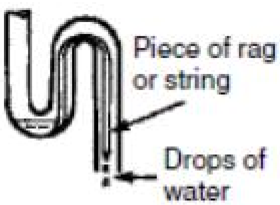
Induced siphonage

Back pressure: compression occurs due to resistance to flow at the base of a stack. The positive pressure displaces water in the lowest trap. Causes are a too small radius bottom bend, an undersized stack or the lowest branch fitting too close to the base of the stack



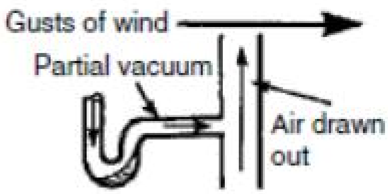
Back pressure or compression

• **Capillary action**: a piece of rag, string or hair caught on the trap outlet.



Capillary attraction

• **Wavering out**: gusts of wind blowing over the top of the stack can cause a partial vacuum to disturb water seals.



Wavering out

Through evaporation: If a plumbing fixture or drain is not used frequently, or during prolonged periods of inactivity, the water within the trap can slowly evaporate due to exposure to air. As the water level decreases, the seal is compromised, and the trap becomes ineffective in blocking sewer gases. This is common in traps associated with floor drains, sinks, or other fixtures that may not see regular use. In situations where traps are not used regularly, periodic checks and adding water manually can prevent the loss of the water seal due to evaporation.

4) SANITARY PIPE WORKS.

There are 5 types of drainage water pipes used in building construction:

a) **Soil Pipes -** A soil pipe is designed to carry soiled water from the toilet, urinal to the sewer. Soil pipes are vented high at the top or near to the top of a building, thanks to soil pipe stacks, to allow gases produced by waste to vent safely into the atmosphere.

b) **Waste Pipes -** It is often a smaller diameter pipe that carries waste water from your sinks, washing machine, shower, bath and any other appliance you may have that uses water. It can be narrower than a soil pipe as the waste pipe is only designed to carry water. It also doesn't need the same venting system as a soil pipe since it does not generate harmful gas.

c) **Vent Pipe -** Vent pipes are attached to the top of soil and waste pipe for the release of bad odors.

d) **Rainwater Pipes -** These pipes are attached to the roof or open area above building for the removal or collection of rainwater. The rainwater pipes are drawn to the ground level in case of removal or these are connected to the rainwater collection network or tank for rainwater harvesting. These are generally made of PVC material.

e) Anti-Siphonage Pipes - These are connected to the outlets of toilets which are provided to maintain water seal to prevent entry of foul gases of the sewer lines into the toilets and bathrooms. These are made of PVC and their sizes must conform to the respective standard specifications.

5) PLUMBING SYSTEMS

Plumbing is the system or network of pipes, tanks, fittings, and other fixtures used inside a building for supplying water and removing water-borne wastes. Plumbing system follows the general law of nature gravity, and pressure.

Following are the four principle systems adopted in plumbing work in building

✓ Two pipe system.

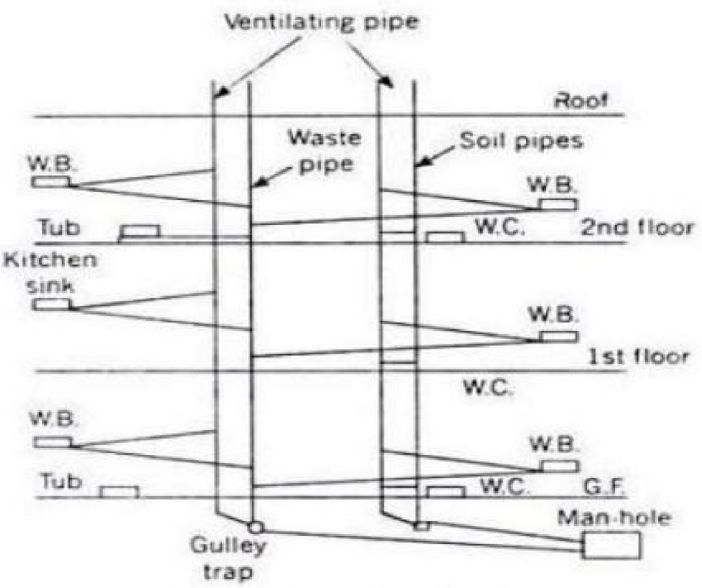
✓ One pipe system.

✓ Single stack system

✓ Partially ventilated single stack system

Two pipe system: -

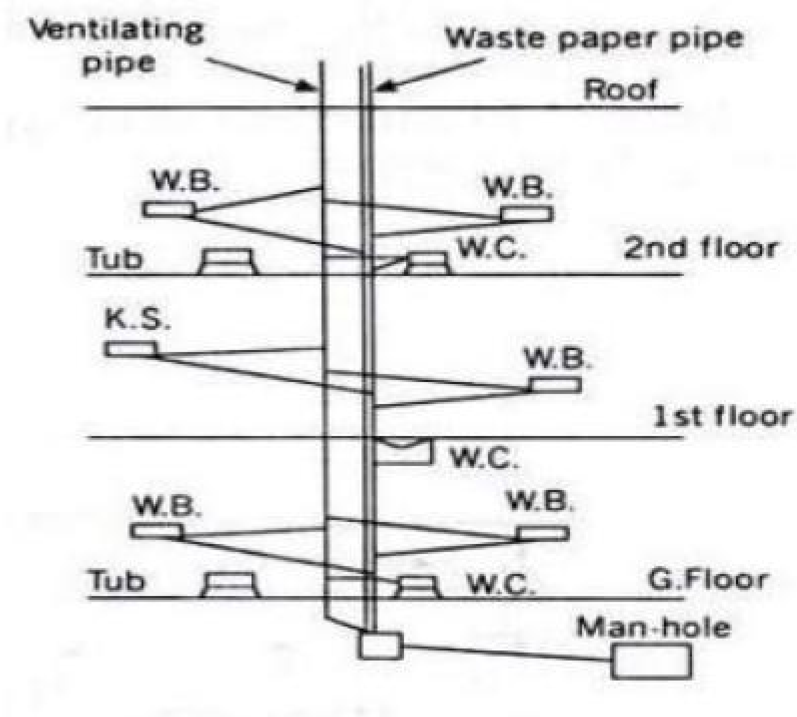
1. This is the best and most improved type of system of plumbing.
2. In this system, two sets of vertical pipes are laid, i.e. one for draining soil and other for draining sullage.
3. The pipe of the first set carrying soil are called soil pipes. and the pipes of the second set carrying sullage from baths etc. are called sullage pipe or waste pipe
4. The soil fixtures, such as latrines and urinals are thus all connected through branch pipes to the vertical pipe.
5. Where the sludge fixtures such as baths, sinks, wash-basins, etc are all connected through branch pipes to the vertical waste pipe.
6. The soil pipe as well as the waste pipe are separately ventilated by providing separate vent pipe.



Two-pipe system.

One pipe system: -

1. In this system, instead of using two separate pipes (for carrying sullage and soil, as it done in the two-pipe system), only main vertical pipe is provided which collects the soil as well as the sullage water from their respective fixtures through the branch pipes.
2. This main pipe is ventilated in itself by providing cowl at its top and in addition to this, a separate vent pipe is also provided.

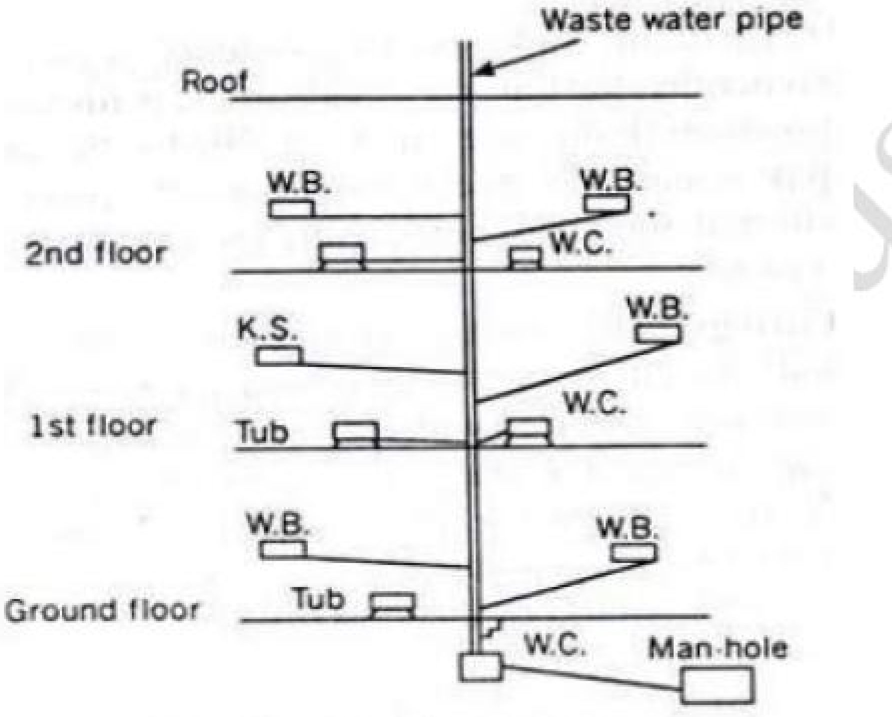


One-pipe system.

Single Stack System: -

1. This system is a single pipe system without providing any separate ventilation pipe.

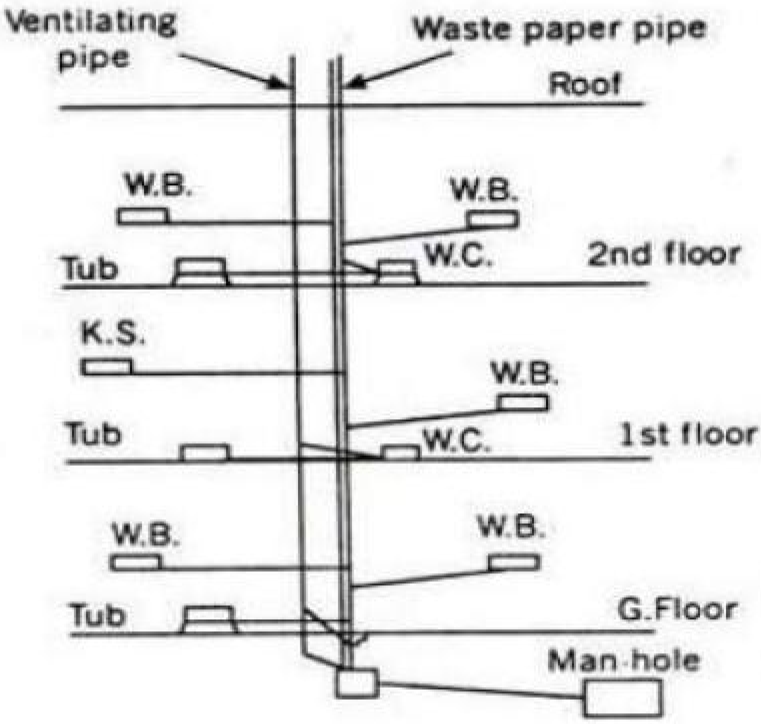
2. It uses only one pipe which carries the sewage as well as sullage, and is not provided with any separate vent pipe, except that it itself is extended upto about 2m higher than the roof level and provided with a cowl for removal of foul gases



Single stack system.

Partially ventilated single stack: -

1. This is an improved form of single stack system in the sense that in this system, the traps of water closets are separately ventilated by a separate vent pipe called relief vent pipe.
2. This system uses two pipes as in single pipe system but the cost of branches is considerably reduced compared to single pipe system.



One-pipe partially ventilated system